Leela Prasanna P

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SUMMARY

I am a dedicated and enthusiastic individual with a strong desire to build a successful career in a progressive and growth-oriented environment. I believe in continuous learning and strive to enhance my skills to adapt to evolving challenges. My goal is to apply my knowledge, creativity, and determination to deliver impactful results while maintaining a commitment to excellence and integrity. I value teamwork, innovation, and responsibility, and aim to contribute meaningfully to both organizational success and the betterment of the community around me.

EDUCATION

Bachelor of Technology in Computer Enineering(AIML) **2023-2027**

MLR Institute of Technology

Intermediate **2021 - 2023**

Narayana Junior College(66.00%)

School **2020 - 2021**

Pragathi Central School (60 %)

PROJECTS

Smart Traffic Signal Management Using Real-Time Vehicle Detection

- Developed an AI-powered Smart Traffic Signal System using YOLOv8, OpenCV, and PyTorch for real-time vehicle detection and tracking.
- Implemented a threshold-based decision mechanism to count only vehicles that have passed through the signal, eliminating false triggers from parked or idle vehicles.
- Designed an automated traffic signal control module that dynamically switches signals based on live traffic density, enhancing road traffic flow efficiency.
- Optimized the system for real-time performance with lightweight YOLOv8n architecture, achieving high accuracy with minimal computational cost.
- Built a modular and easily deployable architecture using Python 3.10, ensuring compatibility and scalability for future enhancements (e.g., multi-lane tracking, live camera integration).
- Processed and analyzed traffic footage to demonstrate adaptive signal timing, reducing unnecessary idle time and congestion.
- Applied computer vision, object detection, and automation principles to simulate real-world traffic management scenarios.

AI-Powered Healthcare Diagnostic System

- Developed a convolutional neural network (CNN) using ResNet18 to classify Chest X-ray images into Normal and Pneumonia categories, achieving high diagnostic accuracy.
- Applied Transfer Learning on pretrained ImageNet weights, enabling faster convergence and improved performance on a limited healthcare dataset.
- Implemented data preprocessing pipelines using PyTorch & OpenCV, including resizing, normalization, and data augmentation (rotation, flipping, brightness adjustment) to enhance model robustness.
- Performed end-to-end model training and evaluation on Kaggle GPU (T4), efficiently handling large image batches and optimizing runtime.

- Monitored model learning using accuracy and loss curves, and validated results with confusion matrix, classification report, and ROC-AUC score.
- Visualized and interpreted model predictions on test X-rays, demonstrating clear distinction between infected and normal lungs through prediction heatmaps.
- Experimented with different architectures (ResNet, EfficientNet) and tuning hyperparameters (learning rate, optimizer, batch size) to achieve optimal results.
- Documented the entire workflow — dataset handling, model training, and evaluation — ensuring full reproducibility and scalability.
- Explored potential deployment using Flask/Streamlit for an interactive web-based diagnostic interface.
- Showcased how AI can assist radiologists in early pneumonia or COVID-19 detection, emphasizing real-world healthcare impact.

SKILLS

Web: HTML, CSS, JavaScript, Node JS.

Databases:SQL

Programming languages :C, N8N , Android Studio ,Python, Java(oops)

PERSONALITY TRAITS & SKILLS:

- Communication Skills.
- Composed in high-stress situations.
- Passionate about learning new things.

CERTIFICATES

- Cisco Certification
- IBM Certification
- AWS Certificate

ACHIVEMENTS

- Partcipated in SAWIT LLM in Language Model Training.
- Partcipated in Finathon by Aczen .
- Participated in AWS cloud workshop .
- participated in SIH 2025